Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2025

Course: B.Tech | *Time: 03 Hours*
Code: 119406 | *Subject: Material ,Testing & Evaluation* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) The following material has maximum ductility?
(i) Mild steel (iii) Nickel
(ii) Copper (iv) Aluminium

(b) The hardness is the property of a material due to which it
(i) can be drawn into wires (iii) can cut another metal
(ii) breaks with little permanent distortion (iv) can be rolled or hammered into thin sheets

(c) Bitumen felt
(i) is used as water proofing material (iii) is made from bitumen and hessian fibres
(ii) is used as damp proofing material (iv) all the above

(d) Elastic deformation is
(i) Permanent deformation (iii) Reversible deformation
(ii) Non-reversible deformation (iv) Failure due to fatigue

(e) The commonly used colour pigment in paints is
(i) ambers (iii) iron oxide
(ii) carbon black (iv) lamp black

(f) Which of the following machine is used to measure compressive strength?
(i) Universal testing machine (iii) Fatigue testing machine
(ii) Impact testing machine (iv) Erichsen machine

(g) Which of the following device is used to measure hardness in Mohs?
(i) Rockwell tester (iii) Universal testing machine
(ii) Sclerometer (iv) Gyro meter

(h) Plastic deformation in metals is mainly caused by
(i) Grain growth (iii) Chemical reactions
(ii) Dislocation movement (iv) Elastic bending

(i) Which of the following materials is predominated by thermal failure?
(i) Actual (iii) PMMA
(ii) Nylon (iv) Polycarbonate

(j) The AISI-SAE designation system is commonly used for classifying
(i) Nonferrous metals (iii) Cast irons
(ii) Stainless steels (iv) Ceramics

Q.2 (a) Explain the difference between M-Sand and river sand in terms of properties and applications. **[7]**

(b) Evaluate the advantages and disadvantages of using structural steel over concrete in high-rise buildings. **[7]**

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Q.3	(a)	Describe the tensile test and explain its significance in material testing.	[7]
	(b)	Compare and contrast brittle and ductile fractures with suitable examples.	[7]
Q.4	(a)	How do bending and torsion tests help in understanding the mechanical behavior of materials?	[7]
	(b)	Discuss structural integrity assessment, and it importance.	[7]
Q.5	(a)	Discuss the importance of force measurement in testing civil engineering materials. Also List the various types of bituminous materials used in road construction.	[7]
	(b)	Explain the concept of creep in materials and its fundamental characteristics.	[7]
Q.6	(a)	Discuss the temperature transition approach for brittle fracture of steel.	[7]
	(b)	Suggest an appropriate hardness test for a polymer material and justify your choice.	[7]
Q.7	(a)	Describe briefly at least four types of structural steel used in civil engineering.	[7]
	(b)	Discuss the importance of standard testing procedure in material evaluation.	[7]
Q.8	(a)	Explain AiSI -SAE steel designation system with examples.	[7]
	(b)	Discuss fatigue failure with S N Curve.	[7]
Q.9	(a)	Evaluate the effectiveness of Non Destructive testing in Quality Control.	[7]
	(b)	What are forces responsible for causing natural defects in Timber. Discuss method adopted for its protection?	[7]

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